

Counterexamples to Erdős Problem 341

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Abstract

Erdős problem 341 asks whether the consecutive gap sequence of the pair-sum-avoiding greedy extension of every finite seed set is eventually periodic. We give a negative answer by exhibiting the following counterexample

$$A = \{1, 2, 3, 5, 7, 13, 22, 27, 28, 32, 36, 40, 47, 48, 52, 63, 71, 77, 81, 89, 97\}.$$

The proof constructs an infinite set S satisfying the exact greedy recurrence $t \in S \iff t \notin S + S$ for $t > 97$. We show that S is the greedy extension of the 21-element seed set A , and use a scale-eight sumset controller embedded in three residue classes modulo 49 and a finite modular shield to prove that the consecutive gap sequence of S is not eventually periodic. Motivated by this counterexample, we further obtain an infinite family of counterexamples by a slight modification of the construction.

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1 Introduction and the main theorem

Write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. For $X, Y \subseteq \mathbb{N}_0$ and $c \in \mathbb{N}_0$, put

$$X + Y = \{x + y \mid x \in X, y \in Y\}, \quad c + X = \{c + x \mid x \in X\}, \quad X^c = \mathbb{N}_0 \setminus X.$$

Equal summands are allowed in every sumset. For integers $a \leq b$, write $[a, b]_{\mathbb{Z}} = \{a, a + 1, \dots, b\}$.

Let $A = \{a_1 < \dots < a_k\} \subseteq \mathbb{N}$ be a nonempty finite set. For $n \geq k$, define

$$a_{n+1} = \min\{t \in \mathbb{N} \mid t > a_n, t \notin A_n + A_n\}, \quad A_n = \{a_1, \dots, a_n\}. \quad (1)$$

Let $d_m = a_{m+1} - a_m$ denote the consecutive gaps. Erdős and Graham asked whether, for every finite initial set A , the resulting gap sequence $(d_m)_{m \geq 1}$ is eventually periodic. The question appears in Erdős and Graham [1, p. 53] and is catalogued as Erdős Problem 341 in Bloom's online database [2]. Related periodicity criteria are due to Calkin and Finch [3], while Calkin and Erdős [4] proved that a natural class of aperiodic sum-free sets is incomplete. We answer the arbitrary finite-seed question negatively.

Theorem 1.1. *Let*

$$A = \{1, 2, 3, 5, 7, 13, 22, 27, 28, 32, 36, 40, 47, 48, 52, 63, 71, 77, 81, 89, 97\}.$$

For the sequence generated from A by (1),

$$\forall K \geq 1 \ \forall h \geq 1 \ \exists m \geq K \quad d_{m+h} \neq d_m.$$

Thus its consecutive gap sequence is not eventually periodic.

Lemma 1.2 (Exact recurrence implies the greedy rule). *Let $S \subseteq \mathbb{N}$ be infinite, let $N \in S$, and put $A = S \cap [1, N]_{\mathbb{Z}}$. Suppose that*

$$t \in S \iff t \notin S + S \quad (t > N). \quad (2)$$

Then the increasing enumeration of S is exactly the greedy extension of A .

Proof. Scan the integers $t > N$ in order. Inductively, immediately before testing t , the selected set is $S \cap [1, t-1]_{\mathbb{Z}}$. Since all elements of S are positive, every representation $t = u + v$ with $u, v \in S$ has $u, v < t$, and hence

$$t \in (S \cap [1, t-1]_{\mathbb{Z}}) + (S \cap [1, t-1]_{\mathbb{Z}}) \iff t \in S + S.$$

Thus (2) makes the rule select exactly the elements of S , completing the induction. $\square$

2 A scale-eight sumset controller

We use the standard identity

$$[a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} = [a + c, b + d]_{\mathbb{Z}} \quad (a \leq b, c \leq d). \quad (3)$$

Define

$$L = \bigcup_{k \geq 0} [8^k, 4 \cdot 8^k - 1]_{\mathbb{Z}} \quad (4)$$

$$Y_0 = \{0\} \cup \bigcup_{k \geq 0} [4 \cdot 8^k, 8 \cdot 8^k - 1]_{\mathbb{Z}}, \quad (5)$$

$$Y_1 = [1, 3]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [16 \cdot 8^k - 1, 32 \cdot 8^k - 1]_{\mathbb{Z}}, \quad (6)$$

$$Y_2 = [0, 1]_{\mathbb{Z}} \cup [7, 15]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [64 \cdot 8^k - 1, 128 \cdot 8^k - 1]_{\mathbb{Z}}. \quad (7)$$

At each scale, the corresponding blocks of L and $Y_0 \setminus \{0\}$ partition $[8^k, 8^{k+1} - 1]_{\mathbb{Z}}$. Thus

$$Y_0 = L^c. \quad (8)$$

Lemma 2.1 (Controller identities). *The sets in (5)–(7) satisfy*

$$Y_1 = (Y_0 + Y_0)^c, \quad (9)$$

$$Y_2 = (Y_1 + Y_1)^c, \quad (10)$$

$$Y_0 = \{0\} \cup (1 + (Y_2 + Y_2)^c). \quad (11)$$

Proof. A direct interval calculation gives

$$Y_0 + Y_0 = \{0\} \cup \bigcup_{k \geq 0} [4 \cdot 8^k, 16 \cdot 8^k - 2]_{\mathbb{Z}} = Y_1^c, \quad (12)$$

$$Y_1 + Y_1 = [2, 6]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [16 \cdot 8^k, 64 \cdot 8^k - 2]_{\mathbb{Z}} = Y_2^c, \quad (13)$$

$$Y_2 + Y_2 = [0, 2]_{\mathbb{Z}} \cup [7, 30]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [64 \cdot 8^k - 1, 256 \cdot 8^k - 2]_{\mathbb{Z}} = L - 1. \quad (14)$$

Equations (14) and (8) yield (11). $\square$

Lemma 2.2. *The membership indicator of Y_0 is not ultimately periodic.*

Proof. Suppose that there are $N \geq 0$ and $h \geq 1$ such that

$$n \in Y_0 \iff n + h \in Y_0 \quad (n \geq N).$$

Choose $a = 8^k > \max\{N, h\}$ and put $m = 4a - h$. Then $N < a \leq m \leq 4a - 1$. Hence $m \in L$, so $m \notin Y_0$, whereas $m + h = 4a \in [4a, 8a - 1]_{\mathbb{Z}} \subseteq Y_0$. This contradicts the assumed periodicity. $\square$

3 A finite shield modulo 49

All residues in this section lie in $\mathbb{Z}/49\mathbb{Z}$ and are represented in $\{0, \dots, 48\}$. Put

$$P = \{7, 14, 28\}, \quad B = \{3, 22, 32, 40, 48\}, \quad D = \{1, 2, 5, 13, 27, 36, 47\}, \quad Q = P \cup B. \quad (15)$$

Lemma 3.1 (Finite shield). *In $\mathbb{Z}/49\mathbb{Z}$,*

$$(B + B) \cup (P + B) \cup (D + B) = \mathbb{Z}/49\mathbb{Z} \setminus Q, \quad (16)$$

$$Q \cap ((B + B) \cup (P + B) \cup (D + B) \cup (D + P)) = \emptyset, \quad (17)$$

$$(P + P) \cap Q = P. \quad (18)$$

Moreover, the only ordered pairs in $P \times P$ whose sum lies in Q are

$$7 + 7 = 14, \quad 14 + 14 = 28, \quad 28 + 28 = 49 + 7. \quad (19)$$

Proof. All assertions follow by direct computation in $\mathbb{Z}/49\mathbb{Z}$. $\square$

We now define the infinite set

$$\begin{aligned} S = & D \cup \{49q + b \mid q \in \mathbb{N}_0, b \in B\} \\ & \cup \{49q + 7 \mid q \in Y_0\} \cup \{49q + 14 \mid q \in Y_1\} \cup \{49q + 28 \mid q \in Y_2\}. \end{aligned} \quad (20)$$

Hence

$$S \cap [1, 97]_{\mathbb{Z}} = D \cup B \cup (49 + B) \cup \{7, 28, 63, 77\} = A. \quad (21)$$

Theorem 3.2. *For every $t > 97$,*

$$t \in S \iff t \notin S + S. \quad (22)$$

Proof. Write $t = 49q + r$, with $0 \leq r < 49$. Since $t > 97$, we have $q \geq 2$.

First let $r \in Q = B \cup P$. The six possible pair-sum types are

$$D + D, \quad D + B, \quad D + P, \quad B + B, \quad B + P, \quad P + P.$$

A $D + D$ sum is at most $94 < t$. Equation (17) excludes the next four types from Q , while (18) and (19) leave only the three diagonal $P + P$ pairs. Hence

$$\begin{aligned} 49q + 14 \in S + S & \iff q \in Y_0 + Y_0, \\ 49q + 28 \in S + S & \iff q \in Y_1 + Y_1, \\ 49q + 7 \in S + S & \iff q - 1 \in Y_2 + Y_2. \end{aligned}$$

Indeed, the corresponding representations are

$$\begin{aligned}(49u + 7) + (49v + 7) &= 49(u + v) + 14, \\ (49u + 14) + (49v + 14) &= 49(u + v) + 28, \\ (49u + 28) + (49v + 28) &= 49(u + v + 1) + 7.\end{aligned}$$

The exclusion of every other source type proves the converse implication in each equivalence. The controller identities (9)–(11) therefore make every P -rail integer represented exactly when it is absent from S . Every B -rail integer belongs to S , and the same exhaustive source analysis shows that none is represented. Thus (22) holds for $r \in Q$.

Now let $r \notin Q$. Such an integer t is not in S : its residue excludes every infinite rail, and $t > 97$ excludes the finite set D . By (16), there exist $b \in B$ and $x \in B \cup D \cup P$ such that $x + b \equiv r \pmod{49}$. Set

$$e_x = x \quad (x \in B \cup D), \quad e_7 = 7, \quad e_{14} = 63 = 49 + 14, \quad e_{28} = 28.$$

Every e_x belongs to S . Write

$$e_x = 49u_x + x, \quad x + b = 49c + r.$$

Here $u_x, c \in \{0, 1\}$. Since $q \geq 2$, the quotient $q - u_x - c$ is nonnegative, and

$$t = e_x + (49(q - u_x - c) + b) \in S + S.$$

The second summand is a positive element of a full B -rail. Both summands are positive and sum to t , so each is smaller than t . This completes the proof. $\square$

By (21), Theorem 3.2, and Lemma 1.2, the increasing enumeration of S is exactly the greedy extension of the seed A in Theorem 1.1.

4 Nonperiodicity of the gaps

Lemma 4.1. *Let $T = \{t_1 < t_2 < \dots\} \subseteq \mathbb{N}$ be infinite. If the consecutive gaps $t_{m+1} - t_m$ are eventually periodic, then the membership indicator of T is ultimately periodic.*

Proof. Suppose that the gaps have period $h \geq 1$ from index K onward. Let

$$H = \sum_{j=0}^{h-1} (t_{K+j+1} - t_{K+j}) > 0.$$

Every block of h consecutive gaps in the periodic tail has sum H , so $t_{m+h} = t_m + H$ for $m \geq K$. Since T is infinite, it is unbounded. Thus, given $n \geq t_K$, there is a unique $m \geq K$ such that $t_m \leq n < t_{m+1}$, and

$$t_{m+h} \leq n + H < t_{m+h+1}, \quad (n + H) - t_{m+h} = n - t_m.$$

Thus $n = t_m$ if and only if $n + H = t_{m+h}$, which proves

$$n \in T \iff n + H \in T \quad (n \geq t_K).$$

This is ultimate periodicity of the membership indicator. $\square$

Proof of Theorem 1.1. The residue-7 rail of S satisfies

$$49q + 7 \in S \iff q \in Y_0 \quad (q \in \mathbb{N}_0). \quad (23)$$

Suppose, for contradiction, that membership in S were ultimately periodic with period $h \geq 1$. Then $49h$ would also be an ultimate period. For all sufficiently large q , equation (23) would give

$$q \in Y_0 \iff 49q + 7 \in S \iff 49(q + h) + 7 \in S \iff q + h \in Y_0,$$

contradicting Lemma 2.2. Thus the membership indicator of S is not ultimately periodic. The contrapositive of Lemma 4.1 now shows that the consecutive gaps in the increasing enumeration of S are not eventually periodic, which is the assertion of the theorem. $\square$

5 An infinite family of counterexamples

We further give an infinite family of counterexamples by replacing the finite shield above with a parameterized family of finite shields.

Fix an integer $p \geq 18$, put $M = 7p$, and work in $\mathbb{Z}/M\mathbb{Z}$, represented by $\{0, \dots, M-1\}$. Define

$$\begin{aligned} P_p &= \{p, 2p, 4p\}, & B_p &= \{M-3, M-1, 6p+3\}, & Q_p &= P_p \cup B_p, \\ F_p &= Q_p \cup (Q_p - B_p) \cup (Q_p - P_p), & D_p &= \mathbb{Z}/M\mathbb{Z} \setminus F_p, \end{aligned} \quad (24)$$

where $Q_p - B_p = \{q - b : q \in Q_p, b \in B_p\}$, and similarly for $Q_p - P_p$. Set

$$A_p = D_p \cup B_p \cup (M + B_p) \cup \{p, 4p, M + 2p, M + 4p\}. \quad (25)$$

Theorem 5.1. *For every integer $p \geq 18$, the seed A_p in (25) has $7p - 20$ elements. For the sequence generated from A_p by (1),*

$$\forall K \geq 1 \ \forall h \geq 1 \ \exists m \geq K \quad d_{m+h} \neq d_m.$$

Thus its consecutive-gap sequence is not eventually periodic. In particular, the sets A_p , $p \geq 18$, form an infinite family of counterexamples.

All residues in this section lie in $\mathbb{Z}/M\mathbb{Z}$, with the notation of (24).

Lemma 5.2 (Uniform finite shield). *In $\mathbb{Z}/M\mathbb{Z}$, for every $p \geq 18$,*

$$D_p + B_p = \mathbb{Z}/M\mathbb{Z} \setminus Q_p, \quad (26)$$

$$Q_p \cap ((B_p + B_p) \cup (P_p + B_p) \cup (D_p + B_p) \cup (D_p + P_p)) = \emptyset, \quad (27)$$

$$(P_p + P_p) \cap Q_p = P_p. \quad (28)$$

Moreover, the only ordered pairs in $P_p \times P_p$ whose sum lies in Q_p are

$$p + p = 2p, \quad 2p + 2p = 4p, \quad 4p + 4p = M + p. \quad (29)$$

Proof. Expanding the definition in (24) gives

$$\begin{aligned} F_p &= \{0, 2, \ p-6, p-4, p, p+1, p+3, \ 2p-3, 2p, 2p+1, 2p+3, \\ &\quad 3p-3, 3p-1, 3p, \ 4p, 4p+1, 4p+3, \\ &\quad 5p-3, 5p-1, 5p, 5p+3, \\ &\quad 6p-3, 6p-1, 6p, 6p+3, 6p+4, 6p+6, \ M-3, M-2, M-1\}. \end{aligned} \quad (30)$$

For $p \geq 18$, these are 30 distinct representatives in $\{0, \dots, M-1\}$. In particular, $|D_p| = M-30$. The definition of F_p immediately gives

$$(D_p + B_p) \cap Q_p = \emptyset, \quad (D_p + P_p) \cap Q_p = \emptyset. \quad (31)$$

We prove the reverse inclusion in (26). Let $r \in \mathbb{Z}/M\mathbb{Z}$, and put $f = r + 1$. If $f \notin F_p$, then $f \in D_p$ and

$$r = f + (M-1) \in D_p + B_p.$$

It remains to consider $f \in F_p$. Put

$$G_p = \{0, p+1, 2p+1, 4p+1, 6p+4, M-2\}, \quad E_p = \{p-6, 3p-3, 5p-3, 6p-3, M-3\}.$$

If $f \in G_p$, then direct subtraction shows that $r = f - 1 \in Q_p$. If $f \in F_p \setminus (G_p \cup E_p)$, comparison with (30) shows that $f + 2 \notin F_p$, and hence

$$r = (f + 2) + (M-3) \in D_p + B_p.$$

For the five values of f in E_p , the corresponding representations are

$$\begin{aligned} p-7 &= (2p-10) + (6p+3), & 3p-4 &= (4p-7) + (6p+3), \\ 5p-4 &= (6p-7) + (6p+3), & 6p-4 &= (M-7) + (6p+3), \\ M-4 &= (p-7) + (6p+3). \end{aligned}$$

Thus every residue outside Q_p belongs to $D_p + B_p$, and (31) proves (26).

Direct calculation shows that $B_p + B_p$ and $P_p + B_p$ are disjoint from Q_p . Together with (31), this proves (27). Finally, the three mixed sums of elements of P_p are $3p, 5p, 6p$, all outside Q_p , while the diagonal sums are those in (29). This proves (28) and the ordered-pair assertion. $\square$

We now define the infinite set

$$\begin{aligned} S_p &= D_p \cup \{Mq + b \mid q \in \mathbb{N}_0, b \in B_p\} \\ &\cup \{Mq + p \mid q \in Y_0\} \cup \{Mq + 2p \mid q \in Y_1\} \cup \{Mq + 4p \mid q \in Y_2\}. \end{aligned} \quad (32)$$

The initial controller memberships give

$$S_p \cap [1, 2M-1]_{\mathbb{Z}} = D_p \cup B_p \cup (M + B_p) \cup \{p, 4p, M+2p, M+4p\} = A_p. \quad (33)$$

In particular, $2M-1 = M + (M-1) \in A_p$. The four sets in (33) are disjoint, and therefore

$$|A_p| = (M-30) + 3 + 3 + 4 = M-20 = 7p-20. \quad (34)$$

Theorem 5.3. *For every $p \geq 18$ and every $t > 2M-1$,*

$$t \in S_p \iff t \notin S_p + S_p. \quad (35)$$

Proof. Write $t = Mq + r$, with $0 \leq r < M$. Since $t > 2M-1$, we have $q \geq 2$. The argument is the same as in the proof of Theorem 3.2, with $M, Q_p, P_p, B_p, D_p, S_p$ replacing $49, Q, P, B, D, S$ respectively, and Lemma 5.2 replacing Lemma 3.1.

The only change occurs when $r \notin Q_p$. Such t is not in S_p . By (26), there exist $d \in D_p$ and $b \in B_p$ such that

$$d + b = Mc + r \quad (c \in \{0, 1\}).$$

Since $q \geq 2$, we have $q - c \geq 1$, and

$$t = d + (M(q-c) + b) \in S_p + S_p.$$

This completes the proof. $\square$

By (33), Theorem 5.3, and Lemma 1.2, the increasing enumeration of S_p is exactly the greedy extension of A_p .

Proof of Theorem 5.1. Equation (34) proves the asserted cardinality. The proof of nonperiodicity is the same as in Theorem 1.1. Indeed, suppose, for contradiction, that membership in S_p were ultimately periodic with period $h \geq 1$. Then for all sufficiently large q ,

$$q \in Y_0 \iff Mq + p \in S_p \iff M(q + h) + p \in S_p \iff q + h \in Y_0,$$

contradicting Lemma 2.2. Lemma 4.1 therefore shows that the consecutive gaps in the greedy extension of A_p are not eventually periodic. $\square$

References

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